

# Cloud FPGA Design — Iteration 2 Update

Design Delta from V1 · Orchestration Layer Migration

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**Document type:** Design delta from V1  
**Applies to:** CLOUD\_FPGA\_Design\_V1.pdf  
**Scope:** Orchestration layer migration

## 1 Summary

Iteration 2 relocates the entire orchestration stack — including the build server — from the host machine to the DigitalOcean Droplet. The host is reduced to a minimal hardware agent responsible only for USB/JTAG access and FPGA LAN bridging. All business logic, job management, and compute now run on the Droplet.

## 2 What Changes

### 2.1 Droplet (was: nginx reverse proxy → now: full orchestration node)

The Droplet absorbs everything previously running on the host, in addition to its existing nginx role:

- **FastAPI** API server moves to the Droplet, bound to the public HTTPS interface via nginx
- **Redis** for job queues, FPGA state, session state, and result storage moves to the Droplet (localhost only)
- **Workers**  $\times 10$  move to the Droplet; they issue commands to the host agent over Tailscale rather than acting on hardware directly
- **Build server** (Amaranth → Verilog → Yosys → nextpnr-ecp5 → ecppack pipeline) runs entirely on the Droplet

The Droplet plan must be upgraded accordingly. The Basic shared-CPU tier is insufficient for running nextpnr place-and-route. A **General Purpose** plan (minimum 2 dedicated vCPUs, 8 GB RAM) is required.

### 2.2 Host (was: FastAPI + Redis + Workers + Build server → now: hardware agent only)

The host is stripped to two responsibilities:

1. **JTAG agent** — receives flash commands from the Droplet over Tailscale and executes `openFPGALoader` against the USB hub (`/dev/fpga00–/dev/fpga09`)
2. **Runtime bridge** — receives run payloads from the Droplet, forwards them over UDP/TCP to the target FPGA on the local LAN (`192.168.1.101–.110`), and returns results

The host exposes a minimal HTTP API bound exclusively to its Tailscale IP (`tailscale0`). It has no public surface and holds no state. Redis does not run on the host.

### 2.3 Tailscale Tunnel (was: HTTP proxy channel → now: command channel)

The tunnel carries structured job commands (flash, run, reset) from Droplet workers to the host agent, and results in the return direction. It is no longer a simple HTTP forwarding path.

## 3 Revised Layer Model

| Layer                      | V1                      | V2                                                                    |
|----------------------------|-------------------------|-----------------------------------------------------------------------|
| User                       | Amaranth HDL submission | ← unchanged                                                           |
| Networking / Orchestration | Droplet: nginx only     | Droplet: nginx + FastAPI + Redis + Workers $\times 10$ + Build server |

| Layer             | V1                                              | V2                                              |
|-------------------|-------------------------------------------------|-------------------------------------------------|
| Host              | FastAPI · Redis · Workers<br>×10 · Build server | Hardware agent (JTAG relay +<br>runtime bridge) |
| Hardware Firmware | LiteX SoC per FPGA                              | ← unchanged                                     |
| Physical          | 10× Lattice ECP5-5G                             | ← unchanged                                     |

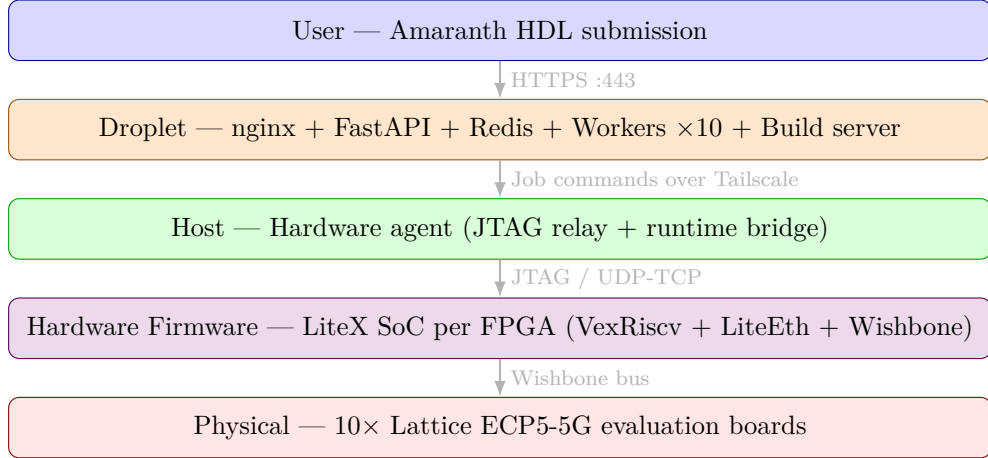


Figure 1: Revised four-layer architecture (V2)

## 4 Revised Runtime Datapath (abbreviated)

| Step | From            | To             | Notes                                            |
|------|-----------------|----------------|--------------------------------------------------|
| 1    | User            | Droplet nginx  | HTTPS :443 — unchanged                           |
| 2    | Droplet worker  | Host agent     | Job command over Tailscale (flash / run / reset) |
| 3    | Host agent      | FPGA           | UDP/TCP on LAN — unchanged                       |
| 4    | FPGA            | Host agent     | Result returned over LAN                         |
| 5    | Host agent      | Droplet worker | Result returned over Tailscale                   |
| 6    | Droplet worker  | Redis          | Job marked complete, session TTL renewed         |
| 7    | Droplet FastAPI | User           | JSON response over HTTPS — unchanged             |

Redis never crosses the Tailscale tunnel. The host agent is stateless.

## 5 Key Design Decisions Introduced

**Host agent interface.** A minimal FastAPI instance on the host, bound to the Tailscale IP only, exposes three endpoints: `POST /flash`, `POST /run`, `POST /reset`. No authentication beyond Tailscale network isolation is required in v2.

**Build isolation.** The sandbox for user HDL compilation (Amaranth  $\rightarrow$  bitstream) now runs on the Droplet. The same sandboxing constraints from V1 apply (no network or filesystem access during build). Resource limits via cgroups or similar should be enforced given shared CPU.

**Concurrent build limiting.** The semaphore that limits concurrent `build_and_program` jobs (previously host-side) moves to the Droplet worker pool. The limit should be tuned to the Droplet's vCPU count rather than the host's.

**Latency.** Run jobs now traverse an additional Tailscale hop (Droplet  $\rightarrow$  host  $\rightarrow$  FPGA  $\rightarrow$  host  $\rightarrow$  Droplet). For the SAT-solving workload this is acceptable. If sub-millisecond round-trip latency becomes a requirement in a future iteration, the runtime bridge path should be revisited.

## 6 No Changes

- Physical layer, FPGA node specifications, power, USB/JTAG, and ethernet topology are unchanged
- LiteX SoC, VexRiscv firmware, LiteEth, and Wishbone interface are unchanged
- API surface (endpoints, request/response contracts, session model) is unchanged from the user's perspective
- Firewall rules on the Droplet (HTTPS + HTTP public, SSH admin-only) are unchanged
- FPGA state machine and job types (`build_and_program`, `run`, `reset`) are unchanged

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